

EXP NO: 2

JavaScript Form Validation for Registration Page

AIM:

Write JavaScript to validate the following fields of the Registration page.

- a) First Name (Name should contains alphabets and the length Should not be less than 6 characters).
- b) Password (Password should not be less than 6 characters length).
- c) E-mail id (should not contain any invalid and must follow the standard pattern name@domain.com)
- d) Mobile Number (Phone number should contain 10 digits only).
- e) Last Name and Address (should not be Empty).

CODE:

Validate.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      display: flex;
      flex-direction: column;
      align-items: center;
      padding: 20px;
    }

    form {
      background-color: white;
```

```
padding: 20px;
padding-right: 30px;
border-radius: 10px;
box-shadow: 0 0 10px rgba(0,0,0,0.1);
width: 320px;
}
```

```
label {
  font-weight: bold;
}
```

```
input, textarea {
  width: 100%;
  margin-top: 5px;
  margin-bottom: 10px;
  padding: 8px;
  border: 1px solid #ccc;
  border-radius: 6px;
}
```

```
.error-message {
  color: red;
  font-size: 13px;
  margin-bottom: 10px;
}
```

```
input[type="submit"] {
  background-color: #4CAF50;
  color: white;
  cursor: pointer;
  font-weight: bold;
  border: none;
}
```

```
input[type="submit"]:hover {
  background-color: #45a049;
}
```

```
    h1 {
        margin-bottom: 20px;
    }
</style>
</head>
<body>
    <h1>Registration Form</h1>
    <form name="RegForm" onsubmit="return validateForm()">
        <label for="firstname">First Name:</label>
        <input type="text" id="firstname" name="firstname" required>
        <span id="firstname-error" class="error-message"></span>

        <label for="lastname">Last Name:</label>
        <input type="text" id="lastname" name="lastname" required>
        <span id="lastname-error" class="error-message"></span>

        <label for="password">Password:</label>
        <input type="password" id="password" name="password" required>
        <span id="password-error" class="error-message"></span>

        <label for="email">Email:</label>
        <input type="text" id="email" name="email" required>
        <span id="email-error" class="error-message"></span>

        <label for="phone">Phone Number:</label>
        <input type="tel" id="phone" name="phone" required>
        <span id="phone-error" class="error-message"></span>

        <label for="address">Address:</label>
        <textarea id="address" name="address" required></textarea>
        <span id="address-error" class="error-message"></span>

        <input type="submit" value="Submit">
    </form>

    <script>
```

```
function validateForm() {  
    let isValid = true;  
  
    let firstname = document.getElementById("firstname").value.trim();  
    let lastname = document.getElementById("lastname").value.trim();  
    let password = document.getElementById("password").value;  
    let email = document.getElementById("email").value.trim();  
    let phone = document.getElementById("phone").value.trim();  
    let address = document.getElementById("address").value.trim();  
  
    const firstnameError = document.getElementById("firstname-error");  
    const lastnameError = document.getElementById("lastname-error");  
    const passwordError = document.getElementById("password-error");  
    const emailError = document.getElementById("email-error");  
    const phoneError = document.getElementById("phone-error");  
    const addressError = document.getElementById("address-error");  
  
    firstnameError.textContent = "";  
    lastnameError.textContent = "";  
    passwordError.textContent = "";  
    emailError.textContent = "";  
    phoneError.textContent = "";  
    addressError.textContent = "";  
  
    if (!/^[A-Za-z]{6,}$/.test(firstname)) {  
        firstnameError.textContent = "First Name must be alphabetic and at least 6 characters.";  
        isValid = false;  
    }  
  
    if (lastname === "") {  
        lastnameError.textContent = "Last Name should not be empty.";  
        isValid = false;  
    }  
}
```

```

    }

    if (password.length < 6) {
        passwordError.textContent = "Password must be at least 6 characters.";
        isValid = false;
    }

    const emailPattern = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
    if (!emailPattern.test(email)) {
        emailError.textContent = "Invalid email format (e.g., name@domain.com).";
        isValid = false;
    }

    if (!/^\d{10}$/.test(phone)) {
        phoneError.textContent = "Phone number must be exactly 10 digits.";
        isValid = false;
    }

    if (address === "") {
        addressError.textContent = "Address should not be empty.";
        isValid = false;
    }

    if (isValid) {
        alert("Registered successfully!");
    }

    return isValid;
}
</script>
</body>
</html>

```

OUTPUT:

The screenshot shows the Eclipse IDE with the 'Form_Validata.html' file open. The Project Explorer on the left shows the project structure, including 'src/main/java' and 'src/main/resources'. The main editor displays the HTML code for the registration form, which includes a title 'Registration Form', a registration form with fields for first name, last name, email, password, phone number, and address, and a submit button. The code is as follows:

```
1<html>
2<head>
3<title>Registration Form</title>
4<style>
5
6    .error-message {
7        color: red;
8        font-size: 14px;
9        margin-top: 5px;
10    }
11    input {
12        padding: 5px;
13        border-radius: 5px;
14        display: block;
15    }
16</style>
17<body style="display: flex; flex-direction: column; justify-content: center; align-items: center;">
18<div>REGISTRATION FORM</div>
19<form name="RegForm" onsubmit="return validateForm()">
20
21    <div>
22        <input type="text" id="first-name" name="first-name" placeholder="Enter First Name" required/>
23        <span id="first-name-error" class="error-message"></span>
24    </div>
25
26    <div>
27        <input type="text" id="last-name" name="last-name" placeholder="Enter Last Name" required/>
28        <span id="last-name-error" class="error-message"></span>
29    </div>
30
31    <div>
32        <input type="text" id="email" name="email" placeholder="Enter Email" required/>
33        <span id="email-error" class="error-message"></span>
34    </div>
35
36    <div>
37        <input type="password" id="password" name="password" placeholder="Enter Password" required/>
38        <span id="password-error" class="error-message"></span>
39    </div>
40
41    <div>
42        <input type="text" id="phone" name="phone" placeholder="Enter Phone Number" required/>
43        <span id="phone-error" class="error-message"></span>
44    </div>
45
46    <div>
47        <input type="text" id="address" name="address" placeholder="Enter Address" required/>
48        <span id="address-error" class="error-message"></span>
49    </div>
50
51    <input type="submit" value="Submit" name="Submit" />
52</form>
53</body>
54</html>
```

The screenshot shows the Eclipse IDE with the 'Form_Validata.html' file open. The Project Explorer on the left shows the project structure, including 'src/main/java' and 'src/main/resources'. The main editor displays the JavaScript code for the registration form, which includes a function 'validateForm' that checks the first name, password, email, phone number, and address fields. The code is as follows:

```
65
66
67const passwordError = document.getElementById("password-error");
68const emailError = document.getElementById("email-error");
69const phoneError = document.getElementById("phone-error");
70const lastNameError = document.getElementById("last-name-error");
71const addressError = document.getElementById("address-error");
72
73firstNameError.textContent = "";
74passwordError.textContent = "";
75emailError.textContent = "";
76phoneError.textContent = "";
77lastNameError.textContent = "";
78addressError.textContent = "";
79let isValid = true;
80
81if (/^[a-zA-Z]/.test(firstName)) {
82    firstNameError.textContent = "First Name only contains alphabet";
83    isValid = false;
84}
85else if (firstName.length <= 6) {
86    firstNameError.textContent = "First Name length not less than 6 characters";
87    isValid = false;
88}
89
90if (password.length < 8 || password.length > 30) {
91    passwordError.textContent = "Password must be between 8 and 30 characters.";
92    isValid = false;
93}
94
95const emailPattern = /^[a-zA-Z0-9_+]{1}@[a-zA-Z0-9-]{1}\.[a-zA-Z]{2,}$/;
96if (!emailPattern.test(email)) {
97    emailError.textContent = "Please enter a valid email address.";
98    isValid = false;
99}
100if (phone.length > 10) {
101    phoneError.textContent = "Phone number length must be 10";
102    isValid = false;
103}
104
105if (isValid) {
106    alert("Register successfully");
107    return true;
108}
109else {
110    return false;
111}
112
113</script>
114</body>
115</html>
```

2116231801081

A123431

Registration Form

First Name:
Karthick

Last Name:
S

Password:

Email:
231801079@rajalakshmi.edu.in

Phone Number:
8925751196

Address:
No:2622 2nd Street Ramdass Nagar
Kattupakkam, Chennai-56

Submit

This page says
Registered successfully!

OK

Karthick

Last Name:
S

Password:

Email:
231801079@rajalakshmi.edu.in

Phone Number:
8925751196

Address:
No:2622 2nd Street Ramdass Nagar
Kattupakkam, Chennai-56

Submit

RESULT:

The Registration form with required validation is successfully implemented.